PHYSICAL SCIENCES
Acids and Bases Test

QUESTION 1

1.1 State the Arrhenius definition of an acid and a base. (2)

1.2 State the Lowry-Brønsted definition of an acid and a base. (2)

1.3 Distinguish between a concentrated solution and a dilute solution. (2)

[6]

QUESTION 2

2.1 Classify EACH of the following as a strong or weak acid/base: H_3PO_4, NH_3, $Ba(OH)_2$, $(COOH)_2$ (4)

2.2 Explain, in terms of ionisation, why a strong base produces a higher concentration of OH^- ions than a weak base of the same concentration. (2)

[6]

QUESTION 3

3.1 Calculate how much water must be added to 40 cm^3 of a 0,5 $mol \cdot dm^{-3}$ NaOH solution to change the concentration to 0,05 $mol \cdot dm^{-3}$. (4)

3.2 A stock solution of H_2SO_4 has a concentration of 3 $mol \cdot dm^{-3}$. 20 cm^3 of this stock solution is diluted with water to a final volume of 200 cm^3 (Solution P).

(a) Determine the concentration of Solution P. (3)

(b) 50 cm^3 of Solution P is further diluted with water until its concentration becomes 0,015 $mol \cdot dm^{-3}$ (Solution Q). Calculate the total volume of Solution Q. (3)

[10]

QUESTION 4

4.1 Write a balanced ionisation equation for hydrobromic acid (HBr) reacting with water. (2)

4.2 Write a balanced dissociation equation for potassium hydroxide (KOH) dissolving in water. (2)

[4]

QUESTION 5

5.1 For the reaction below, label the conjugate acid-base pairs (Acid 1, Base 1, Acid 2, Base 2):

$HNO_2(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + NO_2^-(aq)$ (4)

[4]

Goodluck Hlumelo